

2023 年全国硕士研究生招生考试

数学(三) 参考答案

一、选择题

(1)【答案】 A.

【解析】 由偏导数的定义,

$$\left. \frac{\partial f}{\partial x} \right|_{(0,1)} = \lim_{x \rightarrow 0} \frac{f(x,1) - f(0,1)}{x} = \lim_{x \rightarrow 0} \frac{\ln(1+|x \sin 1|)}{x} = \lim_{x \rightarrow 0} \frac{|x|}{x} \sin 1, \text{ 极限不存在.}$$

$$\left. \frac{\partial f}{\partial y} \right|_{(0,1)} = \lim_{y \rightarrow 1} \frac{f(0,y) - f(0,1)}{y-1} = \lim_{y \rightarrow 1} \frac{\ln y}{y-1} = 1. \quad \text{公众号: 旗胜考研}$$

答案选(A).

【评注】 计算偏导数也可用如下方法,如

$$\left. \frac{\partial f}{\partial y} \right|_{(0,1)} = \left. \frac{df(0,y)}{dy} \right|_{y=1} = \left. \frac{d \ln y}{dy} \right|_{y=1} = 1.$$

(2)【答案】 D.

$$\text{【解析】 } \int f(x) dx = \begin{cases} \ln(x + \sqrt{1+x^2}) + C_1, & x \leq 0, \\ (x+1)\sin x + \cos x + C_2, & x > 0. \end{cases}$$

由原函数的连续性知 $C_1 = 1 + C_2$, 令 $C_2 = 0, C_1 = 1$, 则

$$F(x) = \begin{cases} \ln(x + \sqrt{1+x^2}) + 1, & x \leq 0, \\ (x+1)\sin x + \cos x, & x > 0. \end{cases}$$

答案选(D).

(3)【答案】 C.

【解析】 微分方程的特征方程为 $\lambda^2 + a\lambda + b = 0$, 特征根为 $\lambda_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$.

若 $a^2 - 4b > 0$, 则特征方程有两个实根且 $\lambda_1 \neq \lambda_2$.

微分方程的解为 $y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$ 在 $(-\infty, +\infty)$ 上无界.

若 $a^2 - 4b = 0$, 则 $\lambda_1 = \lambda_2 = -\frac{a}{2}$.

微分方程的解为 $y = (C_1 + C_2 x)e^{-\frac{a}{2}x}$ 在 $(-\infty, +\infty)$ 上无界.

若 $a^2 - 4b < 0$, 则 $\lambda_{1,2} = \frac{-a \pm \sqrt{4b - a^2}i}{2}$.

微分方程的解为 $y = e^{-\frac{a}{2}x} \left(C_1 \cos \frac{\sqrt{4b-a^2}}{2}x + C_2 \sin \frac{\sqrt{4b-a^2}}{2}x \right)$.

如果此解在 $(-\infty, +\infty)$ 上有界, 则 $a = 0$, 进而 $b > 0$.

因此答案选(C).

【评注】 本题还可从选项出发, 验证只有(C) 符合条件.

(4)**【答案】** A.

【解析】 由题意知, 正项级数 $\sum_{n=1}^{\infty} (b_n - a_n)$ 收敛, 当然绝对收敛.

当 $\sum_{n=1}^{\infty} a_n$ 绝对收敛时,

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} (b_n - a_n),$$

两个绝对收敛的级数之和绝对收敛, 所以 $\sum_{n=1}^{\infty} b_n$ 绝对收敛,

当 $\sum_{n=1}^{\infty} b_n$ 绝对收敛时, 更多资料关注公众号: 旗胜考研

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n - \sum_{n=1}^{\infty} (b_n - a_n),$$

$\sum_{n=1}^{\infty} a_n$ 也绝对收敛.

答案选(A).

(5)**【答案】** B.

【解析】 (方法一) 分别令(A)(B)(C)(D) 选项中的矩阵为 I_1, I_2, I_3, I_4 .

$$\begin{bmatrix} A & E \\ O & B \end{bmatrix} I_1 = \begin{bmatrix} A & E \\ O & B \end{bmatrix} \begin{bmatrix} |A|B^* & -B^*A^* \\ O & |B|A^* \end{bmatrix} = \begin{bmatrix} |A|AB^* & \cdots \\ \cdots & \cdots \end{bmatrix},$$

不能保证 $|A|AB^* = |A||B|E$, 所以 I_1 不是 $\begin{bmatrix} A & E \\ O & B \end{bmatrix}^*$, 选项(A) 不正确. 同理, 选项(D) 也不正确.

$$\begin{aligned} \begin{bmatrix} A & E \\ O & B \end{bmatrix} I_3 &= \begin{bmatrix} A & E \\ O & B \end{bmatrix} \begin{bmatrix} |B|A^* & -B^*A^* \\ O & |A|B^* \end{bmatrix} \\ &= \begin{bmatrix} |A||B|E & -AB^*A^* + |A|B^* \\ O & |A||B|E \end{bmatrix}, \text{(C) 不正确.} \end{aligned}$$

$$\begin{aligned} \begin{bmatrix} A & E \\ O & B \end{bmatrix} I_4 &= \begin{bmatrix} A & E \\ O & B \end{bmatrix} \begin{bmatrix} |B|A^* & -A^*B^* \\ O & |A|B^* \end{bmatrix} = \begin{bmatrix} |A||B|E & -|A|B^* + |A|B^* \\ O & |A||B|E \end{bmatrix} \\ &= \begin{bmatrix} |A||B|E & O \\ O & |A||B|E \end{bmatrix}, \end{aligned}$$

选项(B) 是正确的.

$$\begin{aligned} \text{(方法二)} \quad \begin{bmatrix} A & E \\ O & B \end{bmatrix}^* &= \begin{bmatrix} A & E \\ O & B \end{bmatrix} \begin{bmatrix} A & E \\ O & B \end{bmatrix}^{-1} = |A||B| \begin{bmatrix} A^{-1} & -A^{-1}B^{-1} \\ O & B^{-1} \end{bmatrix} \\ &= \begin{bmatrix} |A||B|A^{-1} & -|A||B|A^{-1}B^{-1} \\ O & |A||B|B^{-1} \end{bmatrix} = \begin{bmatrix} |B|A^* & -A^*B^* \\ O & |A|B^* \end{bmatrix}. \end{aligned}$$

【评注】 本题用到分块矩阵的逆或伴随, 设 A_{nm}, B_m

$$\begin{aligned}
 \begin{bmatrix} A & C \\ O & B \end{bmatrix}^* &= \begin{bmatrix} A & C \\ O & B \end{bmatrix} \begin{bmatrix} A & C \\ O & B \end{bmatrix}^{-1} = |A| |B| \begin{bmatrix} A^{-1} & -A^{-1}CB^{-1} \\ O & B^{-1} \end{bmatrix} = \begin{bmatrix} |B|A^* & -A^*CB^* \\ O & |A|B^* \end{bmatrix}, \\
 \begin{bmatrix} A & O \\ C & B \end{bmatrix}^* &= \begin{bmatrix} A & O \\ C & B \end{bmatrix} \begin{bmatrix} A & O \\ C & B \end{bmatrix}^{-1} = |A| |B| \begin{bmatrix} A^{-1} & O \\ -B^{-1}CA^{-1} & B^{-1} \end{bmatrix} = \begin{bmatrix} |B|A^* & O \\ -B^*CA^* & |A|B^* \end{bmatrix}, \\
 \begin{bmatrix} C & A \\ B & O \end{bmatrix}^* &= \begin{bmatrix} C & A \\ B & O \end{bmatrix} \begin{bmatrix} C & A \\ B & O \end{bmatrix}^{-1} = (-1)^{nm} |A| |B| \begin{bmatrix} O & B^{-1} \\ A^{-1} & -A^{-1}CB^{-1} \end{bmatrix} \\
 &= (-1)^{nm} \begin{bmatrix} O & |A|B^* \\ |B|A^* & -A^*CB^* \end{bmatrix}, \\
 \begin{bmatrix} O & A \\ B & C \end{bmatrix}^* &= \begin{bmatrix} O & A \\ B & C \end{bmatrix} \begin{bmatrix} O & A \\ B & C \end{bmatrix}^{-1} = (-1)^{nm} |A| |B| \begin{bmatrix} -B^{-1}CA^{-1} & B^{-1} \\ A^{-1} & O \end{bmatrix} \\
 &= (-1)^{nm} \begin{bmatrix} -B^*CA^* & |A|B^* \\ |B|A^* & O \end{bmatrix}.
 \end{aligned}$$

(6)【答案】 B.

【解析】 (方法一) 配方法

$$\begin{aligned}
 &(x_1 + x_2)^2 + (x_1 + x_3)^2 - 4(x_2 - x_3)^2 \\
 &= x_1^2 + 2x_1x_2 + x_2^2 + x_1^2 + 2x_1x_3 + x_3^2 - 4x_2^2 + 8x_2x_3 - 4x_3^2 \\
 &= 2x_1^2 - 3x_2^2 - 3x_3^2 + 2x_1x_2 + 2x_1x_3 + 8x_2x_3 \\
 &= 2\left[\left(x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_3\right)^2 - \frac{1}{4}x_2^2 - \frac{1}{4}x_3^2 - \frac{1}{2}x_2x_3\right] - 3x_2^2 - 3x_3^2 + 8x_2x_3 \\
 &= 2\left(x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_3\right)^2 - \frac{7}{2}x_2^2 - \frac{7}{2}x_3^2 + 7x_2x_3 \\
 &= 2\left(x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_3\right)^2 - \frac{7}{2}(x_2 - x_3)^2
 \end{aligned}$$

正确答案为(B).

(方法二) 合同变换法

二次型对应的对称矩阵 $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & 4 \\ 1 & 4 & -3 \end{bmatrix}$,

$$\begin{aligned}
 \begin{bmatrix} A \\ \dots \\ E \end{bmatrix} &= \begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & 4 \\ 1 & 4 & -3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 0 & -\frac{7}{2} & \frac{7}{2} \\ 0 & \frac{7}{2} & -\frac{7}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 0 & -\frac{7}{2} & 0 \\ 0 & 0 & 0 \\ 1 & -\frac{1}{2} & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{\sqrt{2}}{2\sqrt{7}} & -1 \\ 0 & \frac{\sqrt{2}}{\sqrt{7}} & 1 \\ 0 & 0 & 1 \end{bmatrix}.
 \end{aligned}$$

$$\text{令 } P = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{\sqrt{2}}{2\sqrt{7}} & -1 \\ 0 & \frac{\sqrt{2}}{\sqrt{7}} & 1 \\ 0 & 0 & 1 \end{bmatrix}, x = Py, \text{ 则 } f = y_1^2 - y_2^2.$$

(方法三) 特征值

二次型对应的对称矩阵 $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & 4 \\ 1 & 4 & -3 \end{bmatrix}$.

由 $|\lambda E - A| = \begin{vmatrix} \lambda - 2 & -1 & -1 \\ -1 & \lambda + 3 & -4 \\ -1 & -4 & \lambda + 3 \end{vmatrix} = \lambda(\lambda + 7)(\lambda - 3) = 0$, 得 A 的特征值为 $3, -7, 0$,

故正惯性指数为 1, 负惯性指数为 1, 故选(B).

(方法四) 可逆线性变换

令 $\begin{cases} z_1 = x_1 + x_2, \\ z_2 = x_1 + x_3, \\ z_3 = x_3, \end{cases}$ 则 $f = z_1^2 + z_2^2 - 4(z_1 - z_2)^2 = -3z_1^2 - 3z_2^2 + 8z_1z_2$.

再由配方法、合同变化法或特征值.

二次型对应的矩阵 $A = \begin{bmatrix} -3 & 4 & 0 \\ 4 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, A 的特征值为 $1, -7, 0$, 正惯性指数为 1, 负惯性指数

为 1, 故选(B).

(7)【答案】 D.

【解析】 设 $\gamma = x_1\alpha_1 + x_2\alpha_2 = x_3\beta_1 + x_4\beta_2$, 即

$$x_1\alpha_1 + x_2\alpha_2 - x_3\beta_1 - x_4\beta_2 = 0. \quad (*)$$

下面求解该方程组:

$$[\alpha_1, \alpha_2, -\beta_1, -\beta_2] = \begin{bmatrix} 1 & 2 & -2 & -1 \\ 2 & 1 & -5 & 0 \\ 3 & 1 & -9 & -1 \end{bmatrix} \xrightarrow{\text{行初等变换}} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \text{ (行最简形),}$$

方程组 (*) 同解于 $\begin{cases} x_1 = -3x_4, \\ x_2 = x_4, \\ x_3 = -x_4. \end{cases}$

通解为 $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = x_4 \begin{pmatrix} -3 \\ 1 \\ -1 \\ 1 \end{pmatrix}, x_4 \in \mathbf{R}.$

$$\begin{aligned} \gamma &= x_1\alpha_1 + x_2\alpha_2 = -3x_4 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + x_4 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \\ &= x_4 \begin{pmatrix} -3 \\ -6 \\ -9 \end{pmatrix} + x_4 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = k \begin{pmatrix} 1 \\ 5 \\ 8 \end{pmatrix}, k = -x_4 \in \mathbf{R}. \end{aligned}$$

正确答案为(D).

(8)【答案】 C.

【解析】 由 $X \sim P(1), P\{X = k\} = \frac{1}{k!}e^{-1} (k = 0, 1, 2, \dots), EX = 1$.

$$E(|X - EX|) = E(|X - 1|) = |0 - 1| P\{X = 0\} + \sum_{k=1}^{\infty} |k - 1| P\{X = k\}$$

$$\begin{aligned}
 &= \frac{1}{e} + \sum_{k=1}^{\infty} (k-1)P\{X=k\} \\
 &= \frac{1}{e} + \sum_{k=0}^{\infty} (k-1)P\{X=k\} - (0-1)P\{X=0\} \\
 &= \frac{1}{e} + E(X-1) + \frac{1}{e} = \frac{2}{e}.
 \end{aligned}$$

答案选(C).

(9)【答案】 D.

【解析】 由 $\frac{(n-1)S_1^2}{\sigma^2} \sim \chi^2(n-1)$, $\frac{(m-1)S_2^2}{2\sigma^2} \sim \chi^2(m-1)$, 且相互独立, 得

$$\frac{\frac{(n-1)S_1^2}{\sigma^2}/n-1}{\frac{(m-1)S_2^2}{2\sigma^2}/m-1} = \frac{2S_1^2}{S_2^2} \sim F(n-1, m-1).$$

答案选(D).

(10)【答案】 A.

【解析】 X_1 与 X_2 独立, 服从的分布均为 $N(\mu, \sigma^2)$, 记 $Y = X_1 - X_2$, 则 $Y \sim N(0, 2\sigma^2)$, $\hat{\sigma} = a|Y|$. 由 $E\hat{\sigma} = \sigma$, 即

$$\begin{aligned}
 E(a|Y|) &= \int_{-\infty}^{+\infty} a|y| \frac{1}{\sqrt{2\pi} \sqrt{2\sigma^2}} e^{-\frac{y^2}{4\sigma^2}} dy = \int_0^{+\infty} 2ay \frac{1}{2\sqrt{\pi}\sigma} e^{-\frac{y^2}{4\sigma^2}} dy \\
 &= a \cdot \frac{2\sigma}{\sqrt{\pi}} = \sigma,
 \end{aligned}$$

解得 $a = \frac{\sqrt{\pi}}{2}$. 答案选(A).

二、填空题

(11)【答案】 $\frac{2}{3}$.

【解析】 $\lim_{x \rightarrow \infty} x^2 \left(2 - x \sin \frac{1}{x} - \cos \frac{1}{x} \right)$
 $= \lim_{x \rightarrow \infty} x^2 \left\{ 2 - x \left[\frac{1}{x} - \frac{1}{6x^3} + o\left(\frac{1}{x^3}\right) \right] - \left[1 - \frac{1}{2x^2} + o\left(\frac{1}{x^2}\right) \right] \right\}$
 $= \lim_{x \rightarrow \infty} x^2 \left[\frac{2}{3x^2} + o\left(\frac{1}{x^2}\right) \right] = \frac{2}{3}.$

(12)【答案】 $\frac{\pi}{3}$.

【解析】 由题意知 $\frac{\partial f}{\partial x} = -\frac{y}{x^2+y^2}$, $\frac{\partial f}{\partial y} = \frac{x}{x^2+y^2}$.

$\frac{\partial f}{\partial y} = \frac{x}{x^2+y^2}$ 两端对 y 积分得

$$f(x, y) = \arctan \frac{y}{x} + \varphi(x),$$

进而

$$\frac{\partial f}{\partial x} = \frac{-\frac{y}{x^2}}{1 + \left(\frac{y}{x}\right)^2} + \varphi'(x) = -\frac{y}{x^2+y^2} + \varphi'(x),$$

则 $\varphi'(x) = 0$, 有 $\varphi(x) = C$, 所以 $f(x, y) = \arctan \frac{y}{x} + C$.

由 $f(1, 1) = \frac{\pi}{4}$, 得 $C = 0$. 因此 $f(\sqrt{3}, 3) = \arctan \sqrt{3} = \frac{\pi}{3}$.

(13)【答案】 $\frac{e^x + e^{-x}}{2}$.

【解析】 (方法一) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$,

$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots + \frac{(-1)^n x^n}{n!} + \cdots$,

两式相加, 得

$$e^x + e^{-x} = 2 + 2 \cdot \frac{x^2}{2!} + 2 \cdot \frac{x^4}{4!} + \cdots + 2 \cdot \frac{x^{2n}}{(2n)!} + \cdots,$$

所以

$$\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = \frac{e^x + e^{-x}}{2}.$$

(方法二) 令 $S(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$, 则

$$S'(x) = \sum_{n=1}^{\infty} \frac{x^{2n-1}}{(2n-1)!}, S''(x) = \sum_{n=2}^{\infty} \frac{x^{2n-2}}{(2n-2)!} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = S(x).$$

解二阶常系数微分方程

$$S''(x) - S(x) = 0,$$

满足初始条件 $S(0) = 1, S'(0) = 0$,

$$S(x) = C_1 e^{-x} + C_2 e^x, \text{ 可得 } C_1 = C_2 = \frac{1}{2},$$

$$\text{所以 } \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = \frac{e^x + e^{-x}}{2}.$$

【评注】 实际上本题是双曲余弦函数 $\operatorname{ch} x = \frac{e^x + e^{-x}}{2}$.

(14)【答案】 $2e^t - 2t - 2$.

【解析】 由题设知 $\frac{\int_0^t f(x) dx}{t} = \frac{f(t)}{t} - t$, 则

$$\int_0^t f(x) dx = f(t) - t^2.$$

该式两端对 t 求导得

$$f(t) = f'(t) - 2t,$$

即 $f'(t) - f(t) = 2t$, 解此线性方程得

$$f(t) = e^{\int dt} \left(C + \int 2te^{-\int dt} dt \right) = Ce^t - 2t - 2,$$

由 $f(0) = 0$ 得 $C = 2$, 故 $f(t) = 2e^t - 2t - 2$.

(15)【答案】 8.

【解析】 已知题中方程组有解, 所以 $r(A) = r(B)$,

$$\text{其中 } A = \begin{bmatrix} a & 0 & 1 \\ 1 & a & 1 \\ 1 & 2 & a \\ a & b & 0 \end{bmatrix}, B = \begin{bmatrix} a & 0 & 1 & 1 \\ 1 & a & 1 & 0 \\ 1 & 2 & a & 0 \\ a & b & 0 & 2 \end{bmatrix}.$$

又因为 $\begin{vmatrix} a & 0 & 1 \\ 1 & a & 1 \\ 1 & 2 & a \end{vmatrix} = 4 \neq 0$, 所以 $r(\mathbf{A}) = 3$, 从而 $r(\mathbf{B}) = 3$, $|\mathbf{B}| = 0$.

$$\begin{aligned} |\mathbf{B}| &= \begin{vmatrix} a & 0 & 1 & 1 \\ 1 & a & 1 & 0 \\ 1 & 2 & a & 0 \\ a & b & 0 & 2 \end{vmatrix} \xrightarrow{\text{按 4 列展开}} - \begin{vmatrix} 1 & a & 1 \\ 1 & 2 & a \\ a & b & 0 \end{vmatrix} + 2 \begin{vmatrix} a & 0 & 1 \\ 1 & a & 1 \\ 1 & 2 & a \end{vmatrix} \\ &= 8 - \begin{vmatrix} 1 & a & 1 \\ 1 & 2 & a \\ a & b & 0 \end{vmatrix} = 0. \end{aligned}$$

所以 $\begin{vmatrix} 1 & a & 1 \\ 1 & 2 & a \\ a & b & 0 \end{vmatrix} = 8$.

(16)【答案】 $-\frac{1}{3}$.

【解析】 $(X+Y)$ 与 $(X-Y)$ 相关系数为 $\rho = \frac{\text{Cov}(X+Y, X-Y)}{\sqrt{D(X+Y)} \sqrt{D(X-Y)}}$,

因为 $X \sim B(1, p)$, $DX = p(1-p)$, $Y \sim B(2, p)$, $DY = 2p(1-p)$, 且 X, Y 独立,

$$D(X+Y) = D(X-Y) = DX + DY = p(1-p) + 2p(1-p) = 3p(1-p),$$

$$\begin{aligned} \text{Cov}(X+Y, X-Y) &= \text{Cov}(X, X) + \text{Cov}(X, -Y) + \text{Cov}(Y, X) + \text{Cov}(Y, -Y) = DX - DY \\ &= p(1-p) - 2p(1-p) = -p(1-p), \end{aligned}$$

$$\text{总之 } \rho = \frac{\text{Cov}(X+Y, X-Y)}{\sqrt{D(X+Y)} \sqrt{D(X-Y)}} = \frac{-p(1-p)}{3p(1-p)} = -\frac{1}{3}.$$

三、解答题

(17)【解】 (I) 等式 $ae^x + y^2 + y - \ln(1+x)\cos y + b = 0$ 两端对 x 求导得

$$ae^x + 2yy' + y' - \frac{\cos y}{1+x} + \ln(1+x)y' \sin y = 0.$$

将 $x = 0, y = 0, y'(0) = 0$ 代入以上两式得

$$a + b = 0, a - 1 = 0,$$

由此可得 $a = 1, b = -1$.

(II) 等式 $e^x + 2yy' + y' - \frac{\cos y}{1+x} + \ln(1+x)y' \sin y = 0$ 两端对 x 求导得

$$e^x + 2yy'' + 2y'^2 + y'' + \frac{\cos y}{(1+x)^2} + \frac{y' \sin y}{1+x} + [\ln(1+x)y' \sin y]' = 0.$$

将 $x = 0, y = 0, y'(0) = 0$ 代入以上式得

$$1 + y''(0) + 1 = 0,$$

则 $y''(0) = -2 < 0$, 故 $x = 0$ 为 $y(x)$ 的极大值点.

(18)【解】 (I) 所求面积为

$$\begin{aligned} S &= \int_1^{+\infty} \frac{dx}{x \sqrt{1+x^2}} = \int_1^{+\infty} \frac{dx}{x^2 \sqrt{1+\left(\frac{1}{x}\right)^2}} = - \int_1^{+\infty} \frac{d\frac{1}{x}}{\sqrt{1+\left(\frac{1}{x}\right)^2}} \\ &= -\ln\left(\frac{1}{x} + \sqrt{1+\frac{1}{x^2}}\right) \Big|_1^{+\infty} = \ln(1+\sqrt{2}). \end{aligned}$$

(II) 所求旋转体体积为

$$\begin{aligned} V &= \pi \int_1^{+\infty} \frac{dx}{x^2(1+x^2)} = \pi \int_1^{+\infty} \left(\frac{1}{x^2} - \frac{1}{1+x^2} \right) dx \\ &= -\pi \left(\frac{1}{x} + \arctan x \right) \Big|_1^{+\infty} = \pi \left(1 - \frac{\pi}{4} \right). \end{aligned}$$

【评注】第1问的积分计算还有其他方法.

方法一 令 $t = \sqrt{x^2+1}$, $S = \int_{\sqrt{2}}^{+\infty} \frac{dt}{t^2-1} = \frac{1}{2} \ln \frac{t-1}{t+1} \Big|_{\sqrt{2}}^{+\infty} = \ln(\sqrt{2}+1)$.

方法二 $S = \int_1^{+\infty} \frac{1}{x\sqrt{1+x^2}} dx \xrightarrow{x=\tan t} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sec^2 t}{\tan t \sec t} dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc t dt$
 $= -\ln |\csc t + \cot t| \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \ln(\sqrt{2}+1).$

(19)【解】如图,

设 $D_1 = \{(x, y) \mid x^2 + y^2 \leq 1, (x-1)^2 + y^2 \leq 1\}$,

$D_2 = \{(x, y) \mid x^2 + y^2 > 1, (x-1)^2 + y^2 \leq 1\}$,

则

$$\begin{aligned} & \iint_D |\sqrt{x^2+y^2}-1| dx dy \\ &= \iint_{D_1} (1-\sqrt{x^2+y^2}) dx dy + \iint_{D_2} (\sqrt{x^2+y^2}-1) dx dy \end{aligned}$$

而

$$\iint_{D_1} (1-\sqrt{x^2+y^2}) dx dy = 2 \int_0^{\frac{\pi}{3}} d\theta \int_0^1 (1-r) r dr + 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} (1-r) r dr$$

$$= \frac{\pi}{9} + 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \left(2\cos^2\theta - \frac{8}{3}\cos^3\theta \right) d\theta$$

$$= \frac{\pi}{9} + \frac{\pi}{3} + \sin 2\theta \Big|_{\frac{\pi}{3}}^{\frac{\pi}{2}} - \frac{16}{3} \left(\sin\theta - \frac{1}{3}\sin^3\theta \right) \Big|_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \frac{4}{9}\pi + \frac{3\sqrt{3}}{2} - \frac{32}{9}.$$

$$\iint_{D_2} (\sqrt{x^2+y^2}-1) dx dy = 2 \int_0^{\frac{\pi}{3}} d\theta \int_1^{2\cos\theta} (r-1) r dr$$

$$= 2 \int_0^{\frac{\pi}{3}} \left(\frac{8}{3}\cos^3\theta - 2\cos^2\theta + \frac{1}{6} \right) d\theta = -\frac{5}{9}\pi + \frac{3}{2}\sqrt{3}.$$

所以 $\iint_D |\sqrt{x^2+y^2}-1| dx dy = 3\sqrt{3} - \frac{\pi}{9} - \frac{32}{9}.$

【评注】还可计算 $\iint_D (\sqrt{x^2+y^2}-1) dx dy = \iint_D (\sqrt{x^2+y^2}-1) dx dy - \iint_{D_1} (\sqrt{x^2+y^2}-1) dx dy.$

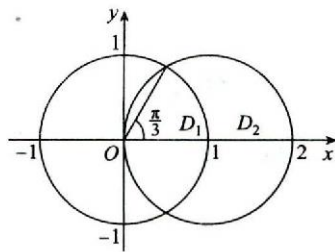
(20)【证明】(I) 由泰勒公式可知

$$f(x) = f(0) + f'(0)x + \frac{f''(\eta)}{2!}x^2 = f'(0)x + \frac{f''(\eta)}{2!}x^2, \text{ 其中 } \eta \text{ 介于 } 0 \text{ 与 } x \text{ 之间.}$$

则

$$f(a) = f'(0)a + \frac{f''(\eta_1)}{2!}a^2, (0 < \eta_1 < a),$$

①



$$f(-a) = -f'(0)a + \frac{f''(\eta_2)}{2!}a^2, (-a < \eta_2 < 0), \quad (2)$$

$$\textcircled{1} + \textcircled{2} \text{ 得 } f(a) + f(-a) = \frac{a^2}{2}[f''(\eta_1) + f''(\eta_2)]. \quad (3)$$

又 $f''(x)$ 在 $[\eta_2, \eta_1]$ 上连续, 则必有最小值 m 和最大值 M . 而

$$m \leq \frac{1}{2}[f''(\eta_1) + f''(\eta_2)] \leq M.$$

由介值定理知, 存在 $\xi \in [\eta_2, \eta_1] \subset (-a, a)$, 使得 $f''(\xi) = \frac{1}{2}[f''(\eta_1) + f''(\eta_2)]$.

$$\text{代入 } \textcircled{3} \text{ 式得 } f''(\xi) = \frac{1}{a^2}[f(a) + f(-a)].$$

(II) 设 $f(x)$ 在 $x_0 \in (-a, a)$ 取得极值, 则 $f'(x_0) = 0$, 由泰勒公式可知

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2!}(x - x_0)^2$$

$$= f(x_0) + \frac{f''(\xi)}{2!}(x - x_0)^2, \text{ 其中 } \xi \text{ 介于 } x_0 \text{ 与 } x \text{ 之间,}$$

$$\text{则 } f(-a) = f(x_0) + \frac{f''(\xi_1)}{2!}(-a - x_0)^2, (-a < \xi_1 < x_0),$$

$$f(a) = f(x_0) + \frac{f''(\xi_2)}{2!}(a - x_0)^2, (x_0 < \xi_2 < a),$$

$$\begin{aligned} |f(a) - f(-a)| &= \left| \frac{f''(\xi_2)}{2!}(a - x_0)^2 - \frac{f''(\xi_1)}{2!}(a + x_0)^2 \right| \\ &\leq \frac{|f''(\xi_2)|}{2}(a - x_0)^2 + \frac{|f''(\xi_1)|}{2}(a + x_0)^2. \end{aligned}$$

又 $|f''(x)|$ 在 $[\xi_1, \xi_2]$ 上连续, 则必有最大值 M .

故存在 $\eta \in [\xi_1, \xi_2] \subset (-a, a)$, 使得 $|f''(\eta)| = M$.

$$|f(a) - f(-a)| \leq \frac{M}{2}[(a - x_0)^2 + (a + x_0)^2]$$

$$\leq \frac{M}{2}(4a^2) = 2a^2 |f''(\eta)|,$$

$$|f''(\eta)| \geq \frac{1}{2a^2} |f(a) - f(-a)|.$$

$$(21) \text{【解析】} (I) A = AE = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}.$$

$$\begin{aligned} (II) \text{ 令 } |A - \lambda E| &= \begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & -1-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{vmatrix} \\ &= \frac{(1+\lambda)C_2 + C_3}{(1+\lambda)C_2 + C_3} \begin{vmatrix} 1-\lambda & 1 & 2+\lambda \\ 2 & -1-\lambda & 1-(1+\lambda)^2 \\ 0 & 1 & 0 \end{vmatrix} \\ &= -(\lambda+2)(\lambda-2)(\lambda+1) = 0. \end{aligned}$$

求得 A 的特征值为 $\lambda_1 = -1, \lambda_2 = -2, \lambda_3 = 2$.

先求解方程组 $(A + E)x = 0$

$$A + E = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{\text{行变换}} \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$(A + E)x = 0$ 的通解为 $x = k_1(-1, 0, 2)^T, k_1 \in \mathbf{R}$.

取 A 的属于 $\lambda_1 = -1$ 的特征向量为 $\xi_1 = (-1, 0, 2)^T$.

再求解 $(A + 2E)x = 0$

$$A + 2E = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{\text{行变换}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

$(A + 2E)x = 0$ 的通解为 $x = k_2(0, 1, -1)^T, k_2 \in \mathbf{R}$.

取 A 的属于 $\lambda_2 = -2$ 的特征向量为 $\xi_2 = (0, 1, -1)^T$.

最后求解 $(A - 2E)x = 0$

$$A - 2E = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -3 & 1 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{\text{行变换}} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix},$$

$(A - 2E)x = 0$ 的通解为 $x = k_3(4, 3, 1)^T, k_3 \in \mathbf{R}$.

取 A 的属于 $\lambda_3 = 2$ 的特征向量为 $\xi_3 = (4, 3, 1)^T$.

$$\text{令 } P = [\xi_1, \xi_2, \xi_3] = \begin{bmatrix} -1 & 0 & 4 \\ 0 & 1 & 3 \\ 2 & -1 & 1 \end{bmatrix}, \text{ 则 } P^{-1}AP = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

(22)【解】(I) X 的分布函数

$$F_X(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^x \frac{e^t}{(1+e^t)^2} dt = \frac{e^x}{1+e^x}, -\infty < x < +\infty.$$

(II) 设 Y 的分布函数为 $F_Y(y)$, 概率密度为 $f_Y(y)$.

(方法一) 定义法

$$F_Y(y) = P\{Y \leq y\} = P\{e^X \leq y\},$$

当 $y < 0$ 时, $F_Y(y) = 0$.

$$\text{当 } y \geq 0 \text{ 时, } F_Y(y) = P\{e^X \leq y\} = P\{X \leq \ln y\} = F(\ln y) = \frac{y}{1+y}.$$

$$\text{故 } f_Y(y) = F'_Y(y) = \frac{1}{(1+y)^2}.$$

$$\text{总之 } f_Y(y) = \begin{cases} \frac{1}{(1+y)^2}, & y \geq 0, \\ 0, & \text{其他.} \end{cases}$$

(方法二) 公式法

$y = e^x$ 在 $(-\infty, +\infty)$ 上单调上升, 值域为 $(0, +\infty)$, 反函数为 $x = \ln y$,

$$f_Y(y) = \begin{cases} f(\ln y)(\ln y)' = \frac{e^{\ln y}}{(1+e^{\ln y})^2} \cdot \frac{1}{y} = \frac{1}{(1+y)^2}, & y > 0, \\ 0, & \text{其他.} \end{cases}$$

(III) $EY = \int_0^{+\infty} \frac{y}{(1+y)^2} dy$ 积分不收敛, 故 Y 的期望不存在.